

Hafsa Y

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Profile Summary

ECE undergraduate with a solid foundation in programming, databases, and software development. Skilled in Python, Kotlin, SQL, and ML frameworks, with experience building practical applications and optimizing solutions. Strong problem-solving and teamwork abilities, with a focus on delivering efficient and scalable systems.

Technical Skills

- **Programming:** Python, C, SQL, Kotlin, HTML, Java(basics)
- **Frameworks/Tools:** Pandas, Matplotlib, Streamlit, Jetpack Compose
- **Databases:** MySQL
- **Soft Skills:** Problem-solving, Communication, Teamwork

Education

- **B.Tech in ECE**, NSS College of Engineering, Palakkad — APJAKTU 2022–2026 (CGPA: 9.12)
- **HSS**, KPRP HSS 2021 (99.5%)
- **SSLC**, KPRP HSS 2019 (96%)

Internships

- **SQL Intern**, Codetech IT Solutions Dec 2024 – Jan 2025
Optimized SQL queries, enhanced backend schema performance, and created analytics reports.
- **Data Science Intern**, Corizo Apr – May 2023
Performed data cleaning, visualization, and built ML models for predictive analytics.

Fellowship

Research Fellow – IASc-INSa-NASi Summer Research Fellowship, IIST May–Jul 2025

- Developed a **privacy-preserving federated learning framework (DP-FedCAT)** for IoT wearables using **Python, PyTorch**, and **Opacus**, integrating **differential privacy** and **knowledge distillation**.

Projects

KTU Study Assistant (Python, Streamlit, Google Generative AI) Jul 2025

- Built an academic assistant with modules like an **AI Chatbot Tutor**, **Quiz Generator**, **SGPA/CGPA Calculator**, **PYQ Finder**, and **Note Summarizer**, integrating the **Gemini 1.5 API** for intelligent query handling and summarization.

TAMGAN-Based S11 Prediction Model (Python, PyTorch) Dec 2025

- Built a TAMGAN-based deep learning model using 1,648 CST/HFSS samples with advanced preprocessing, achieving $R^2 = 0.93$ and $10^6\times$ faster inference compared to SVR, Random Forest, and Gradient Boosting models.

EduTrack App (Kotlin, Jetpack Compose) Apr 2025

- Developed an **Android app** for teacher attendance tracking with an intuitive **Jetpack Compose UI**.

Machine Learning Models (Python, Scikit-learn) Apr 2023 – Mar 2024

- Wine Quality Classifier using Random Forest.
- Stock Price Predictor using Linear Regression.

Database Systems (SQL, MySQL) Aug 2024 – Apr 2025

- Employee Management System and Retail Store Database with CRUD operations, role-based access, and optimized queries.

Other Projects

- Smart Pet Feeder (IoT, ESP32, Voice Module)
- Line Follower Robot (Embedded, 8051 MCU, IR Sensors)

Certifications

- Programming, DSA – IIT Madras Jan–Apr 2025
- Essential Mathematics for ML – IIT Roorkee Jan–Mar 2025
- Deep Learning – IIT Kharagpur Jul–Oct 2024
- Control Systems – IIT Madras Jul–Oct 2024
- Advanced Excel – Internshala Aug 2024

Achievements

- Selected – IASc-INSa-NASi SRF 2025
- Selected – Fluxus Hackathon, IIT Indore 2025
- Finalist – Startup Space, E-Summit 2025

Publications

- Hafsa., et al., “**DP-FedCAT: Federated Human Activity Recognition with CATNet, Differential Privacy, and Knowledge Distillation**,” Submitted to *National Conference on Communications (NCC)*, 2025 — Under Review.
- Hafsa., et al., “**Machine Learning Assisted S11 Prediction for Slotted Patch Antennas Using Transformer-Aware Multi-Branch GAN (TAM-GAN)**,” *Manuscript in Preparation*, 2025.

Positions of Responsibility

- Event Coordinator – Elacsta